

Behavior of Solutions of Burgers' Equation with Nonlocal Boundary Conditions

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In this paper, we discuss the long-time behavior of positive solutions of Burgers' equation $u_t = u_{xx} + \varepsilon uu_x$, $0 < x < 1$, $\varepsilon > 0$, $t > 0$ with the nonlocal boundary condition: $u(0, t) = 0$, $u_x(1, t) + \frac{1}{2}\varepsilon u^2(1, t) = au^p(1, t)(\int_0^1 u(x, t) dx)^q$, where $0 < p < \infty$, $0 < q < \infty$. Criteria for stability are given. Blowup in finite time for some solutions is shown. General results are discussed. © 1994 Academic Press, Inc.

1. INTRODUCTION

In this paper, we consider the following initial nonlocal boundary-value problem:

$$\begin{aligned}
 \text{(A)} \quad & u_t = u_{xx} + (f(u))_x & 0 < x < 1, t > 0, \\
 & u(0, t) = 0 & t > 0, \\
 & u_x(1, t) = g(u(1, t), \bar{u}) & t > 0, \\
 & u(x, 0) = u_0(x) & 0 \leq x \leq 1,
 \end{aligned}$$

where $u_0(x)$ is a nonnegative prescribed function and $\bar{u} = \int_0^1 u(x, t) dx$, f and g are continuously differentiable functions satisfying $f(0) = 0$ and $g(0, v) = 0$. In particular, when $f(u) = \frac{1}{2}\varepsilon u^2$ ($\varepsilon > 0$), $g(u, v) = au^p v^q$ with $p > 0$, $q > 0$, or $g(u, v) = au^p v^q - f(u)$, the above problem reduces to Burgers' equation with nonlocal boundary conditions. Of more interest is the case $g(u, v) = au^p v^q - f(u)$, which, from a physical point of view, represents the flux boundary condition $u_x(1, t) + f(u(1, t)) = au^p(1, t)(\int_0^1 u(x, t) dx)^q$.

In a recent paper [9], Levine presented results concerning the asymptotic behavior of positive solutions of Burgers' equation $u_t = u_{xx} + \varepsilon uu_x$, $0 < x < 1$, $t > 0$, subject to the boundary condition $u(0, t) = 0$, $u_x(1, t) = au^p(1, t)$, $t > 0$ with $a > 0$ and $0 < p < \infty$. He analyzed stability

and instability of the stationary solutions and showed that some solutions blow up in finite time. Recently, Anderson and Park generalized these results in two different directions. In [1], the porous medium equation $u_t = (u^m)_{xx} + (\varepsilon/n)(u^n)_x$ is discussed; while in [14] the case $p < 0$ is considered. Both authors considered the case $u_x + f(u) = au^p$ at $x = 1$ as well.

Matano [13] surveyed the general problems in a strongly order-preserving framework. The resulting theory has wide application but does not include the case in which nonlocal boundary conditions are involved.

On the other hand, in [2, 3, 5, 7], Day, Friedman, and Kawohl obtained results for $u_t = Lu$ with a nonlocal boundary condition: $u|_{\partial\Omega} = \int_{\Omega} f(x) u(x, t) dx$, which for our problem reads $u(0, t) = u(1, t) = \int_0^1 f(x) u(x, t) dx$. Here $Lu = \sum_{i,j=1}^n (a_{ij} u_{,i})_{,j} + \sum_{i=1}^n b_i u_{,i} + cu$ is a uniformly elliptic operator. They established the existence and monotonic decay property of solutions.

Our purpose is to study the existence and monotonicity of solutions and their stable and unstable behavior. In particular, some complete analyses are made for Burgers' equation.

In the second section, we discuss several important properties for corresponding stationary solutions of (A) and characterize the set of stationary solutions of Burgers' equation. In Section 3, we establish the comparison theorem for (A) and discuss the local existence and blowup phenomenon of solutions. Finally, we analyze the stability behavior and give the solution diagrams for Burgers' equation in Section 4. Because of the nonlocal term in the boundary condition, the discussion becomes complicated and certain conventional arguments do not apply. However, all results for the purely local boundary condition with $g(u) = au^p$ are covered. Especially, the comparison principle obtained in this paper can be used in more general problems, and it is interesting to note that the solution diagrams for Burgers' equation with $g(u, v) = au^p v^q - \frac{1}{2} \varepsilon u^2$ are totally different from those with $g(u, v) = au^p v^q$.

2. STATIONARY SOLUTIONS AND THEIR BASIC PROPERTIES

The arguments used herein are extensions of those described in [9]. In this section, we concentrate on the stationary solutions of (A), which solve this problem:

$$\begin{aligned} (A_1) \quad & w''(x) + (f(w))' = 0, \quad 0 < x < 1, \\ & w(0) = 0, \quad w'(1) = g(w(1), \bar{w}). \end{aligned}$$

Here, $\bar{w} = \int_0^1 w(x) dx$.

Without placing a monotonicity condition on f and g , we have

LEMMA 2.1. *Nontrivial solutions of (A_1) are of one sign. Furthermore, every positive solution $w(x)$ of (A_1) satisfies $w'(x) > 0$ on $[0, 1]$.*

Proof. The first assertion follows from the maximum principle. As for the second statement, if $w'(x)$ changed sign on $(0, 1)$, w would have an interior extremum, which cannot happen unless $w(x) \equiv \text{const} = 0$. Since $w'(0) > 0$, it follows that $w'(x) \geq 0$ on $[0, 1]$. Suppose that for some $x_0 \in (0, 1]$, $w'(x) > 0$ in $(0, x_0)$ and $w'(x_0) = 0$. Then $w(x_0) > w(x)$ for x in $(0, x_0)$, and the Hopf boundary point lemma yields $w'(x_0) > 0$, which is a contradiction.

(A_1) may have more than one solution, but adding a further assumption on f , we can find that the solutions are ordered.

LEMMA 2.2. *If $f'(u) \geq 0$ for $u > 0$, $w_1(x)$ and $w_2(x)$ are two positive solutions of (A_1) that satisfy $w_1(1) > w_2(1)$, then $w_1(x) > w_2(x)$ on $(0, 1]$.*

Proof. Assume that the statement is not true. There is then an x_0 nearest 1 such that $w_1(x_0) = w_2(x_0)$ and $w_1(x) > w_2(x)$ for x in $(x_0, 1]$. Hence we can find a least x_1 in $(x_0, 1]$, so that $w'_1(x_1) > w'_2(x_1)$. From the conservation law,

$$w'_1(0) = w'_1(x) + f(w_1(x)) = w'_1(x_1) + f(w_1(x_1))$$

and

$$w'_2(0) = w'_2(x) + f(w_2(x)) = w'_2(x_1) + f(w_2(x_1)),$$

it follows that $w'_1(0) > w'_2(0)$.

Because $w_1(0) = w_2(0) = 0$, we know that $w_1(x) > w_2(x)$ near 0. Suppose that x_2 is the first point, such that $w_1(x) > w_2(x)$ in $(0, x_2)$ and $w_1(x_2) = w_2(x_2)$. From the above identities, we find that $w'_1(x_2) > w'_2(x_2)$. For x in $(0, x_2)$, however,

$$\frac{w_1(x) - w_1(x_2)}{x - x_2} < \frac{w_2(x) - w_2(x_2)}{x - x_2},$$

which implies that $w'_1(x_2) \leq w'_2(x_2)$ —a contradiction.

Although we cannot establish an existence theorem for the solutions of (A_1) , by applying further conditions to f and g , we are sometimes able to obtain the uniqueness of solutions of (A_1) .

THEOREM 2.3. *Let f' be nonnegative and strictly increasing for $u > 0$ and suppose that one of two conditions holds: (i) $g(u_1, v_1)/u_1 \geq g(u_2, v_2)/u_2$ or (ii) $g(u_1, v_1)/v_1 \geq g(u_2, v_2)/v_2$ for $u_1 > u_2 > 0$ and $v_1 > v_2 > 0$. Then, at most, one positive solution of (A_1) exists.*

Proof. Suppose that there are two solutions— $w_1(x)$ and $w_2(x)$ of (A_1) —satisfying $w_1(1) > w_2(1)$. By Lemma 2.2, it follows that $w_1(x) > w_2(x)$ on $(0, 1]$ and that $\bar{w}_1 = \int_0^1 w_1(x) dx > \bar{w}_2 = \int_0^1 w_2(x) dx$. Moreover, we have $w'_1(x) > 0$ and $w'_2(x) > 0$ by Lemma 2.1. Let $R(x) = w'_2(x)/w'_1(x)$, $R'(x) = (f'(w_1) - f'(w_2))/R$, $R > 0$ on $(0, 1]$. Therefore, on $(0, 1)$

$$\frac{w'_2(x)}{w'_1(x)} < \frac{w'_2(1)}{w'_1(1)}.$$

Upon multiplication with $w'_1(x)$ and integration, we find that

$$w_2(x_0) < \frac{w'_2(1)}{w'_1(1)} w_1(x_0), \quad x_0 \in (0, 1),$$

and that

$$w_2(x) - w_2(x_0) < \frac{w'_2(1)}{w'_1(1)} (w_1(x) - w_1(x_0)), \quad x_0 < x < 1;$$

that is,

$$\frac{w'_2(1)}{w'_1(1)} w_1(x) - w_2(x) > \frac{w'_2(1)}{w'_1(1)} w_1(x_0) - w_2(x_0) > 0.$$

Letting $x \rightarrow 1^-$, we have

$$\frac{w'_2(1)}{w'_1(1)} w_1(1) - w_2(1) > 0;$$

hence $w'_2(1)/w'_1(1) > w_2(1)/w_1(1)$.

If (i) holds,

$$\frac{w'_2(1)}{w'_1(1)} = \frac{g(w_2(1), \bar{w}_2)}{g(w_1(1), \bar{w}_1)} = \frac{g(w_2(1), \bar{w}_2)/w_2(1)}{g(w_1(1), \bar{w}_1)/w_1(1)} \cdot \frac{w_2(1)}{w_1(1)} \leq \frac{w_2(1)}{w_1(1)}.$$

Combining the above two inequalities leads to a contradiction:

$$\frac{w_2(1)}{w_1(1)} < \frac{w'_2(1)}{w'_1(1)} \leq \frac{w_2(1)}{w_1(1)}.$$

On the other hand, from $(w'_2(1)/w'_1(1)) w_1(x) - w_2(x) > 0$, it follows that $\bar{w}_2/\bar{w}_1 < w'_2(1)/w'_1(1)$. If (ii) were valid,

$$\frac{w'_2(1)}{w'_1(1)} = \frac{g(w_2(1), \bar{w}_2)/\bar{w}_2}{g(w_1(1), \bar{w}_1)/\bar{w}_1} \cdot \frac{\bar{w}_2}{\bar{w}_1} \leq \frac{\bar{w}_2}{\bar{w}_1},$$

which is impossible.

Noteworthy is the equivalence between the solutions of (A_1) and those obtained from integral equations.

THEOREM 2.4. *Let $f' > 0$ for $u > 0$. Suppose that $w(x)$ is a positive solution of $(A_1) \in C^2(0, 1) \cap C^1[0, 1]$. Then $w(x)$ satisfies*

$$F(w; \alpha, \beta) = \int_0^{w(x)} \frac{d\sigma}{g(\alpha, \beta) + f(\alpha) - f(\sigma)} = x$$

for $0 \leq x \leq 1$. Here, $\alpha = w(1)$ and $\beta = \int_0^1 w(x) dx$.

Conversely, if $\alpha > 0$ satisfies $F(\alpha; \alpha, \beta) = 1$, positive $w(x)$ satisfies $F(w; \alpha, \beta) = x$ with $\int_0^1 w(x) dx = \beta$, then $w(x)$ is a positive solution of (A_1) .

Proof. The first statement follows by quadrature. We prove the second part. First, we claim that $h(\sigma) = g(\alpha, \beta) + f(\alpha) - f(\sigma) > 0$ for $\sigma \in [0, \alpha]$. Otherwise, suppose that there existed an $x_0 \in (0, 1)$ with $\sigma_0 = w(x_0)$ such that $h(\sigma_0) = 0$. Then $h(\sigma) = h(\sigma) - h(\sigma_0) = f(\sigma_0) - f(\sigma) \approx f'(\sigma_0)(\sigma_0 - \sigma)$, and we would have

$$x_0 = \int_0^{w(x_0)} \frac{d\sigma}{h(\sigma)} = \int_0^{\sigma_0} \frac{d\sigma}{f(\sigma_0) - f(\sigma)} = +\infty,$$

which is a contradiction.

Next we show that $w(x) < \alpha$ for $0 \leq x < 1$. If not, there would be at least a $x_1 \in [0, 1)$ such that $w(x_1) \geq \alpha$. Because $h(\sigma) > 0$,

$$1 = \int_0^x \frac{d\sigma}{h(\sigma)} \leq \int_0^{w(x_1)} \frac{d\sigma}{h(\sigma)} = x_1,$$

which cannot happen. This argument implies that $w(1) = \alpha$.

We see that $F(w; \alpha, \beta)$ is twice continuously differentiable in w for $0 \leq w \leq w(1)$. Thus, by the inverse function theorem, $w(x)$ is twice differentiable, and we obtain

$$w'(x) + f(w(x)) = g(w(1), \bar{w}) + f(w(1))$$

and

$$w''(x) + (f(w(x)))' = 0.$$

It follows that $w'(1) = g(w(1), \bar{w})$. Since $h(\sigma) > 0$, $F(w; \alpha, \beta) = 0$ implies $w(0) = 0$. ■

As an application of the above theorem, we characterize positive stationary solutions of Burgers' equation with $f(u) = \frac{1}{2}\varepsilon u^2$ and $g(u, v) = au^p v^q$ or $g(u, v) = au^p v^q - \frac{1}{2}\varepsilon u^2$, where $a, \varepsilon > 0$, $0 < p, q < \infty$.

For problem (A_1) with $g(u, v) = au^p v^q$, let $y = \int_0^x w(s) ds$, $h(y) = u^2(x(y))$, where $x = x(y)$ is treated as a function of y . We obtain $h''(y) + \varepsilon h'(y) = 0$ with $h(0) = 0$.

Therefore, $h = C^2(1 - e^{-\varepsilon y})$ and $w = C(1 - e^{-\varepsilon \int_0^x w(s) ds})^{1/2}$. Let $w(1) = \alpha$, $\bar{w} = \beta$,

$$\alpha = C(1 - e^{-\varepsilon \beta})^{1/2}, \quad (2.1.1)$$

$$\begin{aligned} w'(x) &= \frac{1}{2} C(1 - e^{-\varepsilon \int_0^x w(s) ds})^{-1/2} e^{-\varepsilon \int_0^x w(s) ds} \cdot (\varepsilon w(x)) \\ &= \frac{1}{2} C^2 \varepsilon e^{-\varepsilon \int_0^x w(s) ds}. \end{aligned} \quad (2.1.2)$$

Thus,

$$\frac{1}{2} C^2 \varepsilon e^{-\varepsilon \beta} = a \alpha^p \beta^q. \quad (2.2)$$

From (2.1.1) and (2.2), we obtain

$$\alpha^{p-2} = \frac{\varepsilon}{2a} (e^{\varepsilon \beta} - 1)^{-1} \beta^{-q}. \quad (2.3)$$

Hence,

$$F_1(\alpha, \beta) = \frac{1}{2} \varepsilon \alpha F(\alpha; \alpha, \beta) = \int_0^1 \frac{d\sigma}{(2a/\varepsilon) \alpha^{p-2} \beta^q + 1 - \sigma^2} = \frac{1}{2} \varepsilon \alpha. \quad (2.4)$$

On the other hand, the solution $\tilde{w}(x)$ arising from (2.3) and (2.4) satisfies

$$\begin{aligned} \tilde{w}''(x) + f(\tilde{w}(x))' &= 0, \quad 0 < x < 1, \\ \tilde{w}(0) &= 0, \quad \tilde{w}'(1) = g(\alpha, \beta) \end{aligned}$$

with $a = \tilde{w}(1)$. Repeating an argument similar to that leading to (2.1)–(2.2), we have

$$\alpha^{p-2} = \frac{\varepsilon}{2a} (e^{\varepsilon \tilde{w}} - 1)^{-1} \beta^{-q}.$$

Therefore by combining the above equation with (2.3) we find that $\beta = \tilde{w}$.

If $p = 2$, (2.3) is equivalent to

$$Q(\beta) = \frac{2a}{\varepsilon} (e^{\varepsilon \beta} - 1) = \beta^{-q} = R(\beta), \quad (2.5)$$

which we now demonstrate to have exactly one solution.

$Q(0)=0$, $Q(+\infty)=+\infty$, and $Q'(\beta)>0$, while $R(\beta)\rightarrow+\infty$ as $\beta\rightarrow 0$, $R(+\infty)=0$, and $R'(\beta)<0$. Moreover, $F_1(\alpha, \beta)=F_1(\beta)=\int_0^1 d\sigma/((2a/\varepsilon)\beta^q + 1 - \sigma^2) = \frac{1}{2}\varepsilon\alpha$. Therefore, only one $w(1)$ exists for each ε and a . Thus, when $p=2$, there is one solution of (A_1) with this choice of g .

For $p \neq 2$, substituting (2.3) into (2.4) yields

$$\int_0^1 \frac{d\sigma}{(e^{\varepsilon\beta} - 1)^{-1} + 1 - \sigma^2} = \frac{\varepsilon}{2} \left(\frac{\varepsilon}{2a} \right)^{1/(p-2)} (e^{\varepsilon\beta} - 1)^{-1/(p-2)} \beta^{-q/(p-2)}. \quad (2.6)$$

Let $\gamma^2 = (e^{\varepsilon\beta} - 1)^{-1} + 1$, $\beta = (1/\varepsilon) \ln(\gamma^2/(\gamma^2 - 1))$ on $(1, \infty)$, and

$$\int_0^1 \frac{d\sigma}{\gamma^2 - \sigma^2} = \delta (\gamma^2 - 1)^{1/(p-2)} \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^{q/(2-p)},$$

where $\delta = \varepsilon^{(p+q-1)/(p-2)} 2^{(1-p)/(p-2)} a^{1/(2-p)}$. Set

$$\Phi(\gamma) = (\gamma^2 - 1)^{1/(2-p)} \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^{q/(p-2)} \cdot \int_0^1 \frac{d\sigma}{\gamma^2 - \sigma^2} = \delta. \quad (2.7)$$

We then obtain, on $(1, \infty)$,

$$\Phi(\gamma) = \frac{1}{2} (\gamma^2 - 1)^{1/(2-p)} \gamma^{-1} \ln \left(\frac{\gamma+1}{\gamma-1} \right) \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^{q/(p-2)}.$$

If $p > 2$, set $\lambda = 1/(p-2)$, $\mu = q/(p-2)$,

$$\Phi(\gamma) = \frac{1}{2} (\gamma^2 - 1)^{-\lambda} \gamma^{-1} \ln \left(\frac{\gamma+1}{\gamma-1} \right) \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^\mu.$$

and

$$\Phi'(\gamma) = \frac{1}{2} (\gamma^2 - 1)^{-\lambda-1} \gamma^{-2} \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^\mu K(\gamma),$$

where

$$K(\gamma) = [(-2\lambda - 1)\gamma^2 + 1] \ln \left(\frac{\gamma+1}{\gamma-1} \right) - 2\gamma - 2\mu \ln \left(\frac{\gamma+1}{\gamma-1} \right) / \ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right).$$

Because $K(\gamma) < 0$ on $(1, \infty)$, we have $\Phi'(\gamma) < 0$. We can easily see that $\Phi(\gamma) \rightarrow +\infty$ as $\gamma \rightarrow 1^+$ and that $\Phi(+\infty) = 0$. Therefore, (A_1) has one and only one positive solution for all $\varepsilon > 0$, $a > 0$.

For $p < 2$, we first discuss the case $p = 1$,

$$\begin{aligned}\Phi(\gamma) &= \frac{1}{2} (\gamma^2 - 1) \gamma^{-1} \ln \left(\frac{\gamma + 1}{\gamma - 1} \right) \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^{-q} \\ &= \frac{1}{2} \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^{-q} \Phi_1(\gamma),\end{aligned}$$

where $\Phi_1(\gamma) = (\gamma^2 - 1) \gamma^{-1} \ln((\gamma + 1)/(\gamma - 1))$.

We thus find that

$$\begin{aligned}\Phi'(\gamma) &= q \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^{-q-1} [\gamma(\gamma^2 - 1)]^{-1} \\ &\quad \cdot \Phi_1(\gamma) + \frac{1}{2} \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^{-q} \Phi'_1(\gamma),\end{aligned}$$

where

$$\Phi'_1(\gamma) = \gamma^{-2} [(\gamma^2 + 1) \ln((\gamma + 1)/(\gamma - 1)) - 2\gamma].$$

Letting $\Phi_2(\gamma) = (\gamma^2 + 1) \ln((\gamma + 1)/(\gamma - 1)) - 2\gamma$, we obtain

$$\Phi'_2(\gamma) = 2\gamma \ln((\gamma + 1)/(\gamma - 1)) - \frac{4\gamma^2}{\gamma^2 - 1}$$

and

$$\left(\frac{\Phi'_2(\gamma)}{2\gamma} \right)' = \frac{4}{(\gamma^2 - 1)^2} > 0,$$

because $\Phi'_2(\gamma)/2\gamma \rightarrow 0$ as $\gamma \rightarrow +\infty$, $\Phi'_2(\gamma) < 0$ on $(1, \infty)$. $\Phi_2(\gamma) \sim 2(\gamma^2 + 1)/(\gamma - 1) - 2\gamma$ as $\gamma \rightarrow +\infty$; it follows that $\Phi_2(\gamma) > 0$ and, consequently, that $\Phi'_1(\gamma) > 0$ on $(1, \infty)$, which shows that $\Phi'(\gamma) > 0$. Clearly, $\Phi(\gamma) \rightarrow 0$ as $\gamma \rightarrow 1^+$ and $\Phi(+\infty) = +\infty$. Hence, when $p = 1$, we obtain a unique positive solution.

Next, we examine the case of $1 < p < 2$. Set

$$\Phi(\gamma) = \left\{ \frac{1}{2} (\gamma^2 - 1)^{(p-1)/(2-p)} \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^{-q(2-p)} \right\} \Phi_1(\gamma),$$

where $\Phi_1(\gamma)$ is the same as in the case $p = 1$. Obviously, $\Phi'(\gamma) > 0$ for γ in $(1, \infty)$. Because $\Phi(\gamma) \rightarrow 0$ as $\gamma \rightarrow 1^+$ and $\Phi(\gamma) \rightarrow +\infty$ as $\gamma \rightarrow +\infty$, we have exactly one positive solution in this case, as well.

Finally, we consider the case $0 < p < 1$. To simplify the discussion, we start with $p + q \leq 1$. Set $\lambda = 1/(2-p)$ and $\mu = q/(2-p)$; then, $\lambda > \frac{1}{2}$, and $\lambda + \mu \leq 1$. We let

$$\Phi(\gamma) = \frac{1}{2} (\gamma^2 - 1)^\lambda \gamma^{-1} \ln \left(\frac{\gamma+1}{\gamma-1} \right) \left[\ln \left(\frac{\gamma^2}{\gamma^2-1} \right) \right]^{-\mu} \quad (2.8)$$

and find that

$$\Phi'(\gamma) = \frac{1}{2} (\gamma^2 - 1)^{\lambda-1} \gamma^{-2} \left[\ln \left(\frac{\gamma^2}{\gamma^2-1} \right) \right]^{-\mu} \cdot K(\gamma),$$

where

$$K(\gamma) = [(2\lambda - 1)\gamma^2 + 1] \ln \left(\frac{\gamma+1}{\gamma-1} \right) - 2\gamma + 2\mu \ln \left(\frac{\gamma+1}{\gamma-1} \right) / \ln \left(\frac{\gamma^2}{\gamma^2-1} \right)$$

and

$$\begin{aligned} \frac{1}{2} K'(\gamma) &= \left[(2\lambda - 1) \gamma \ln \left(\frac{\gamma+1}{\gamma-1} \right) - \frac{2(\lambda - \mu) \gamma^2}{\gamma^2 - 1} \right] \\ &\quad - \left[\frac{2\mu \gamma^2}{\gamma^2 - 1} - \frac{2\mu}{\gamma^2 - 1} / \ln \left(\frac{\gamma^2}{\gamma^2-1} \right) \right] \\ &\quad - \left\{ \frac{2\mu}{\gamma^2 - 1} \left[2 \ln \left(\frac{\gamma^2}{\gamma^2-1} \right) - \gamma^{-1} \ln \left(\frac{\gamma+1}{\gamma-1} \right) \right] / \left[\ln \left(\frac{\gamma^2}{\gamma^2-1} \right) \right]^2 \right\} \\ &= K_1(\gamma) - K_2(\gamma) - K_3(\gamma). \end{aligned}$$

We claim that $K_1(\gamma) < 0$ because $\gamma \rightarrow +\infty$, $K_1(\gamma)/2\gamma \sim (\lambda + \mu - 1)/\gamma \leq 0$, and

$$\begin{aligned} \left(\frac{K_1(\gamma)}{2\gamma} \right)' &= -\frac{(2\lambda - 1)}{\gamma^2 - 1} + (\lambda - \mu) \frac{\gamma^2 + 1}{(\gamma^2 - 1)^2} \\ &\geq \frac{1}{(\gamma^2 - 1)^2} [(1 - \lambda - \mu) \gamma^2 + 2(2\lambda - 1)] > 0. \end{aligned}$$

$K_2(\gamma) \geq 0$ follows from the well-known inequality $\ln(1+x) \geq x/(1+x)$, with x replaced by $1/(\gamma^2 - 1)$.

To show that $K_3(\gamma) > 0$, we need only prove that

$$H(\gamma) = 2\gamma \ln \left(\frac{\gamma^2}{\gamma^2-1} \right) - \ln \left(\frac{\gamma+1}{\gamma-1} \right) > 0.$$

We have $H'(\gamma) = 2 \ln(\gamma^2/(\gamma^2 - 1)) - 2/(\gamma^2 - 1)$ and $H'(\gamma) \rightarrow 0$ as $\gamma \rightarrow +\infty$. Moreover, $H''(\gamma) = 4/\gamma(\gamma^2 - 1)^2 > 0$; hence, $H'(\gamma) < 0$ on $(1, \infty)$. Note that

$H(\gamma) \rightarrow 0$ as $\gamma \rightarrow +\infty$, and therefore, $H(\gamma) > 0$ for all $\gamma > 1$. Thus, $K'(\gamma) < 0$ on $(1, \infty)$.

If $p+q=1$, which is equivalent to $\lambda+\mu=1$, we find that $\Phi'(\gamma) > 0$ on $(1, \infty)$ because $K(\gamma) \rightarrow 0$ as $\gamma \rightarrow +\infty$. We also observe that $\lim_{\gamma \rightarrow 1^+} \Phi(\gamma) = 0$ and that $\lim_{\gamma \rightarrow +\infty} \Phi(\gamma) = 1$. Thus, if $\delta \geq 1$, that is, if $a \geq 2^{1-p}$, there is no solution; if $\delta < 1$ ($a < 2^{1-p}$), there is one.

For $p+q < 1$ ($\lambda+\mu < 1$), $K(\gamma)$ becomes positive infinity as γ approaches 1, whereas $K(\gamma) \sim 4(\lambda+\mu-1)\gamma < 0$ as $\gamma \rightarrow +\infty$. Thus, $K(\gamma)$ has exactly one sign change and $\Phi(\gamma)$ increases before decreasing. We can also see that $\lim_{\gamma \rightarrow 1^+} \Phi(\gamma) = 0$ and that $\lim_{\gamma \rightarrow +\infty} \Phi(\gamma) = 0$. Hence, (A_1) has none, one, or two positive solutions, depending on whether $\delta > \Phi_M$, $\delta = \Phi_M$, or $\delta < \Phi_M$, where $\Phi_M = \max_{1 < \gamma < \infty} \Phi(\gamma)$. That is,

$$\begin{aligned}\varepsilon &> (\Phi_M)^{(2-p)/(1-p-q)} 2^{(1-p)/(1-p-q)} a^{1/(p+q-1)}, \\ \varepsilon &= (\Phi_M)^{(2-p)/(1-p-q)} 2^{(1-p)/(1-p-q)} a^{1/(p+q-1)},\end{aligned}$$

or

$$\varepsilon < (\Phi_M)^{(2-p)/(1-p-q)} 2^{(1-p)/(1-p-q)} a^{1/(p+q-1)}.$$

The only remaining case is $p+q > 1$ ($\lambda+\mu > 1$). This time, let $\Phi(\gamma) = [\ln(\gamma^2/(\gamma^2-1))]^{-(p+q-1)/(2-p)}$. $\Phi_0(\gamma)$, where $\Phi_0(\gamma)$ is the same as $\Phi(\gamma)$ for $p+q=1$. We can easily check that $\Phi'(\gamma) > 0$ for all γ in $(1, \infty)$, whereas $\Phi(\gamma) \rightarrow 0$ as $\gamma \rightarrow 1^+$, and $\Phi(\gamma) \rightarrow +\infty$ as $\gamma \rightarrow +\infty$. Thus, there is only one

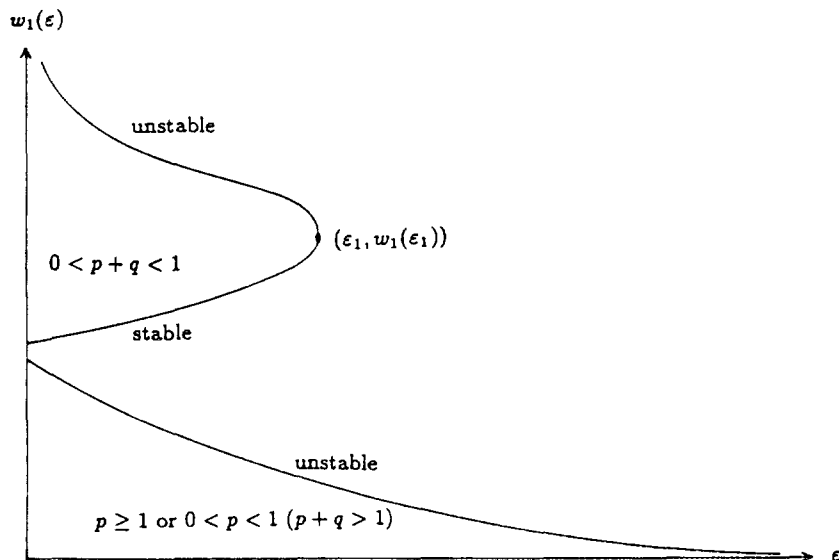


FIG. 1.1. $w_1(\varepsilon)$ for $g(\xi, \eta) = a\xi^p\eta^q$, $a \geq 2^{1-p}$, and $w_1(0) = (2^q/a)^{1/(p+q-1)}$.

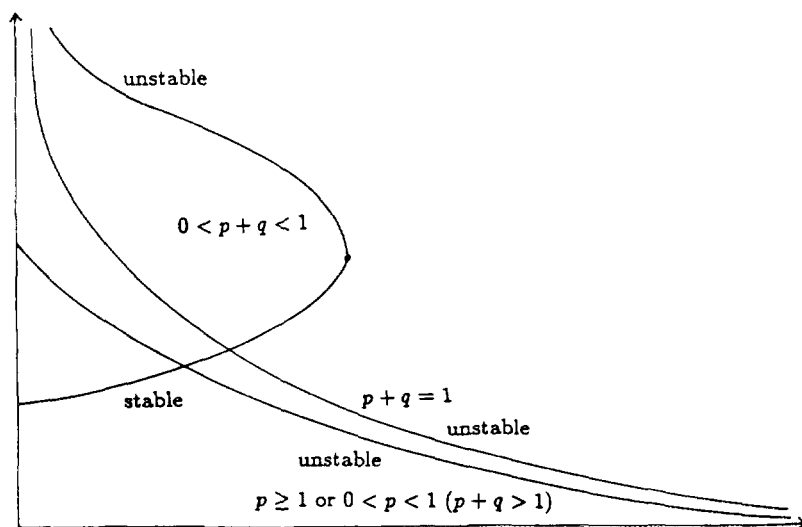


FIG. 1.2. Same as above, except $0 < a < 2^{1-p}$.

positive solution of (A_1) for all $\varepsilon > 0$, $a > 0$. These results are summarized in Figs. 1.1 and 1.2.

Next, we will discuss the problems with $f(u) = \frac{1}{2}\varepsilon u^2$ and $g(u, v) = au^p v^q - \frac{1}{2}\varepsilon u^2$ for (A_1) .

We first obtain

$$\alpha^{p-2} = \left(\frac{\varepsilon}{2a}\right) (1 - e^{-\varepsilon\beta})^{-1} \beta^{-q}. \quad (2.9)$$

Let $\gamma^2 = (1 - e^{-\varepsilon\beta})^{-1}$, then $\beta = 1/\varepsilon \ln(\gamma^2/(\gamma^2 - 1))$, and on $(1, \infty)$,

$$\Phi(\gamma) = \frac{1}{2} \gamma^{2/(2-p)-1} \ln\left(\frac{\gamma+1}{\gamma-1}\right) \left[\ln\left(\frac{\gamma^2}{\gamma^2-1}\right) \right]^{q/(p-2)} = \delta, \quad (2.10)$$

where δ is the same as in (2.7). Note that $\Phi(\gamma)$ is not the same as in (2.7).

If $p > 2$, calculations show that $\Phi'(\gamma) < 0$, $\Phi(1^+) = +\infty$, and $\Phi(+\infty) = 0$; hence, only one solution exists.

The conclusion for $p = 2$ is the same. We only need to show that

$$Q(\beta) = \frac{2a}{\varepsilon} (1 - e^{-\varepsilon\beta}) = \beta^{-q} = R(\beta)$$

has a unique solution.

For $0 < p < 2$, we have

$$\Phi(\gamma) = \frac{1}{2} \gamma^{2\lambda-1} \ln \left(\frac{\gamma+1}{\gamma-1} \right) \left[\ln \left(\frac{\gamma^2}{\gamma^2-1} \right) \right]^{-\mu}, \quad (2.11)$$

where $\lambda = 1/(2-p)$, $\mu = q/(2-p)$;

$$\Phi'(\gamma) = \frac{1}{2} (\gamma^2-1)^{-1} \gamma^{2\lambda-2} \ln \left(\frac{\gamma+1}{\gamma-1} \right) \left[\ln \left(\frac{\gamma^2}{\gamma^2-1} \right) \right]^{-\mu} K(\gamma),$$

with $K(\gamma)$ having the form

$$K(\gamma) = (2\lambda-1)(\gamma^2-1) - 2\gamma / \ln \left(\frac{\gamma+1}{\gamma-1} \right) + 2\mu / \ln \left(\frac{\gamma^2}{\gamma^2-1} \right).$$

Because $2\lambda-1 > 0$ and $\gamma \ln(\gamma^2/(\gamma^2-1)) \leq \ln((\gamma+1)/(\gamma-1))$ on $(1, \infty)$, $K(\gamma) > 0$ if $\mu \geq 1$ —which implies that $p+q \geq 2$. For $p+q > 2$, we find that $\Phi(\gamma) \rightarrow 0$ as $\gamma \rightarrow 1^+$ and $\Phi(\gamma) \rightarrow +\infty$ as γ tends to infinity; there is thus only one solution. If $p+q = 2$, $\lim_{\gamma \rightarrow 1^+} \Phi(\gamma) = \frac{1}{2}$ and $\lim_{\gamma \rightarrow +\infty} \Phi(\gamma) = +\infty$, then, for $\delta \leq \frac{1}{2}$, (A_1) has no solution; but if $\delta > \frac{1}{2}$, there is one solution.

If $p+q \leq 1$, since $\gamma^2-1 \leq 1/\ln(\gamma^2/(\gamma^2-1))$, we easily see that $K(\gamma) < 0$.

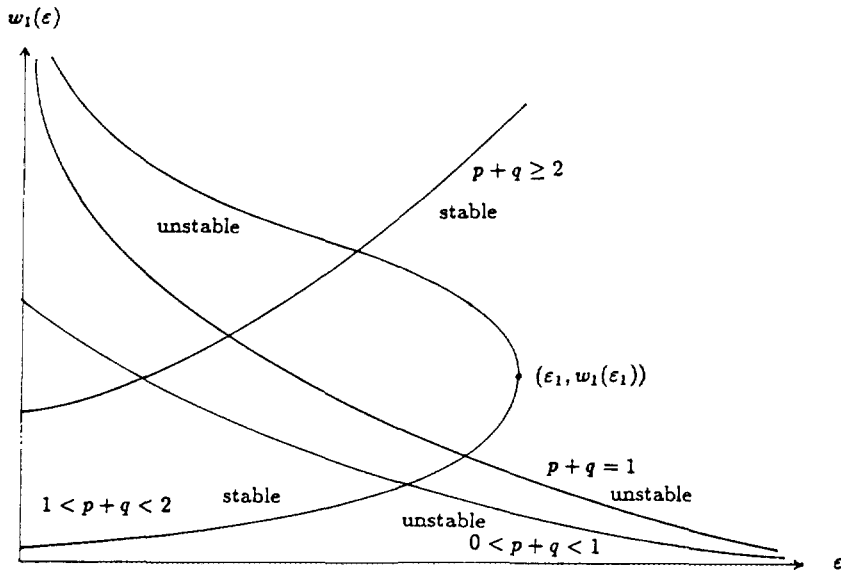
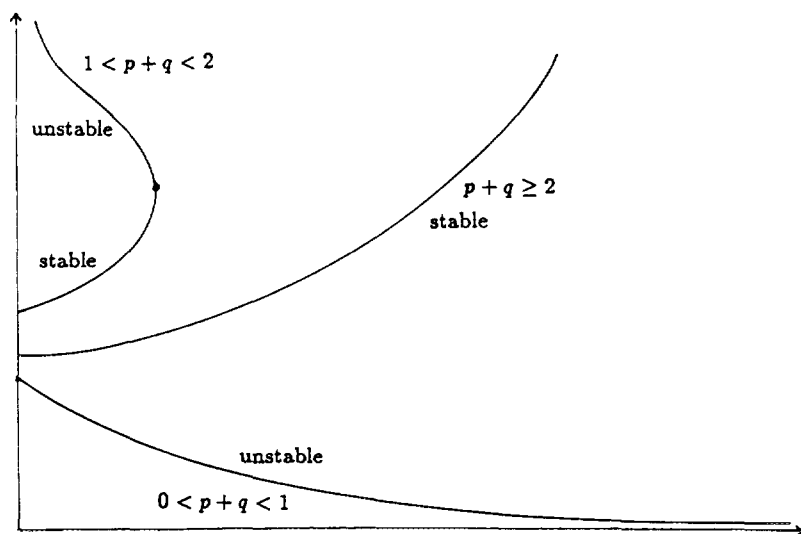


FIG. 2.1. $w_1(\varepsilon)$ for $g(\xi, \eta) = a\xi^p\eta^q - \frac{1}{2}\varepsilon\xi^2$, $a > 2^{1-p}$, and $w_1(0) = (2^q/a)^{1/(p+q-1)}$.

FIG. 2.2. Same as above, except $0 < a \leq 2^{1-p}$.

For $p+q=1$, $\Phi(\gamma) \rightarrow \infty$ as $\gamma \rightarrow 1$ and $\Phi(\gamma) \rightarrow 1$ as $\gamma \rightarrow +\infty$; thus there is one solution if $\delta > 1$ but none if $\delta \leq 1$, whereas for $p+q < 1$, $\Phi(\gamma) \rightarrow \infty$ as $\gamma \rightarrow 1$ and $\Phi(\gamma) \rightarrow 0$ as $\gamma \rightarrow +\infty$, then there is a unique solution.

The case $1 < p+q < 2$ is not so amenable to analysis but, bearing numerical evidence in mind, we may conjecture as follows: Because $\Phi(\gamma) \rightarrow \infty$ as $\gamma \rightarrow 1$ or $+\infty$, there is none, one, or two solutions. See Figs. 2.1, 2.2 for the solution diagrams.

Finally, for $g(u, v) = au^p v^q$ or $g(u, v) = au^p v^q - \frac{1}{2}\epsilon u^2$, if $a < 0$, the null solution is the only nonnegative solution of (A_1) , since by Lemma 2.1, the positive solution must satisfy $w'(1) > 0$, which is impossible for $a < 0$.

3. COMPARISON THEOREM, LOCAL EXISTENCE, AND BLOWUP OF SOLUTIONS

Let $D_T = (0, 1) \times (0, T)$ and $D_T \cup \Gamma_T = [0, 1] \times [0, T]$; by a solution $u(x, t)$ of (A) , we mean that $u(x, t)$ is continuous in $D_T \cup \Gamma_T$, twice continuously differentiable in x , and once in t on D_T . As is well known, comparison theorems are a powerful tool for studying equations of parabolic type. For parabolic problems with nonlocal boundary conditions, however, there are no known comparison theorems, as there are when the conditions are local. For this reason, we attempt to establish such a theory at the beginning of this section. First, we proceed to the definitions of subsolution and supersolution of (A) .

DEFINITION. A function $u(x, t)$ is called a *subsolution* of (A) on D_T if $u \in C^{2,1}(D_T) \cap C(D_T \cup \Gamma_T)$, satisfying

$$\begin{aligned} (A') \quad & u_t \leq u_{xx} + (f(u))_x, & 0 < x < 1, 0 < t < T, \\ & u(0, t) \leq 0, & 0 < t < T, \\ & u_x(1, t) \leq g(u(1, t), \bar{u}), & 0 < t < T, \\ & u(x, 0) \leq u_0(x), & 0 \leq x \leq 1. \end{aligned}$$

A supersolution is defined by (A') with each " $\leq$ " replaced by " $\geq$."

THEOREM 3.1. Suppose that f is continuously differentiable, $g_\xi(\xi, \eta)$ is continuous for $\xi \geq 0, \eta \geq 0$, and $g_\eta(\xi, \eta) \geq 0$ for $\xi \geq 0, \eta \geq 0$. Let u and v be a nonnegative supersolution and a nonnegative subsolution, respectively, of (A). Then $u \geq v$ in $D_T \cup \Gamma_T$.

The proof of this result bears much resemblance to its analog in [4] and is therefore omitted here.

Remark 3.1. For nonnegative subsolution and supersolution, if one of them is positive at $x=1$ for (A), then the conditions on g can be relaxed as follows: $g_\xi(\xi, \eta)$ is continuous for $\xi > 0, \eta > 0$, and $g_\eta(\xi, \eta) \geq 0$ for $\xi > 0, \eta > 0$.

We then obtain the positivity of solutions with nonnegative initial values.

COROLLARY 3.2. Let f be continuously differentiable and suppose that $g_\xi(\xi, \eta)$ is continuous in a neighborhood of $\xi=0$. If $u_0(x) \geq 0$ on $[0, 1]$, then the solution of (A) is positive in $D_T \cup \Gamma_T$, except at $x=0$.

By requiring several restrictions on the initial datum, we also have the following monotonicity result.

COROLLARY 3.3. Suppose that the hypotheses for f and g in Theorem 3.1 are satisfied. If $u_0'' + (f(u_0))' \geq 0$ (≤ 0) on $[0, 1]$, $u_0(0)=0$ and $u_0'(1)=g(u_0(1), \bar{u}_0)$. Then $u_t(x, t) \geq 0$ (≤ 0) in D_T .

Proof. Since the corner compatibility condition is satisfied, by Theorem 7.4 in [8, Chap. V], the function $u_t(x, t)$ is continuous up to the parabolic boundary. Let $v = u_t$. We then have

$$\begin{aligned}
v_t &= v_{xx} + f'(u)v_x + u_x f''(u)v, & 0 < x < 1, 0 < t < T, \\
v(0, t) &= 0, & 0 < t < T, \\
v_x(1, t) &= g_u(u, \bar{u})v + g_{\bar{u}}(u, \bar{u})\bar{v}, & 0 < t < T, \\
v(x, 0) &\geq 0, & 0 \leq x \leq 1.
\end{aligned}$$

Following the same procedure as in Theorem 3.1, we obtain $v \geq 0$ on D_T .

Next, we turn our attention to the local existence theorem for the solutions of (A). To prove the solvability of (A), we need an a priori estimate for $\sup_{D_T} u(x, t)$.

LEMMA 3.4. *Let $u(x, t)$ be the solution of the following problem:*

$$\begin{aligned}
\text{(C)} \quad u_t &= u_{xx} + (f(u))_x, & 0 < x < 1, 0 < t < T, \\
u(0, t) &= 0, & 0 < t < T, \\
u_x(1, t) &= h(t), & 0 < t < T, \\
u(x, 0) &= u_0(x) \geq 0, & 0 \leq x \leq 1,
\end{aligned}$$

where $h(t)$ is a given continuous function. Then for any $M > 0$, with $\max_{0 \leq x \leq 1} u_0(x) < M/2$, there is a T_0 , so that $\sup_{\overline{D_{T_0}}} u(x, t) \leq M$.

Proof. Let $f_{1M} = \sup_{|\xi| \leq M} |f'(\xi)|$, $h_M = \sup_{0 \leq t \leq T} |h(t)|$, and set

$$w(x) = \delta \left[\frac{1}{2} + \frac{1}{\pi} \tan^{-1} \left(\frac{\pi(x-1)}{\delta} \right) \right],$$

where $0 < \delta < M$ and $w(x)$ is taken from [10].

A straightforward computation shows that $0 < w(x) < \delta$, $0 < w'(x) \leq 1$, $0 \leq w''(x) \leq 3/\delta$ in $[0, 1]$ and $w'(1) = 1$.

Choose $\delta = \min\{M, 1/f_{1M}, M/4h_M\}$ and let $v(x, t) = (4h_M/\delta)t + h_M w(x) + M/2$. Then in D_{T_0} with $T_0 = \delta M/16h_M$, $v(x, t) \leq M$, and $v(x)$ is a supersolution of (C). By a conventional comparison theorem, we see that $u(x, t) \leq v(x, t) \leq M$.

We now adopt the monotone iteration scheme to construct the solution of (A).

THEOREM 3.5. *Assume that f is twice continuously differentiable, $g_\xi(\xi, \eta)$ is continuous for $\xi \geq 0$ and $\eta \geq 0$, $g_\eta(\xi, \eta) \geq 0$ for $\xi \geq 0$, $\eta \geq 0$, and $u_0(x) \geq 0$ on $[0, 1]$. Then there is a unique positive solution of (A) on some interval $[0, T_0)$, where $T_0 = T(u_0) \leq \infty$.*

Proof. We set up a sequence $\{u_k\}_{k=1}^\infty$ by the following procedure: Let $u_1 \equiv 0$ and u_{k+1} ($k = 1, 2, \dots$) satisfy the problem

$$\begin{aligned} (C') \quad & (u_{k+1})_t = (u_{k+1})_{xx} + (f(u_{k+1}))_x, & 0 < x < 1, 0 < t < T, \\ & (u_{k+1})(0, t) = 0, & 0 < t < T, \\ & (u_{k+1})_x(1, t) = g(u_k(1, t), \bar{u}_k), & 0 < t < T, \\ & (u_{k+1})(x, 0) = u_0(x), & 0 \leq x \leq 1. \end{aligned}$$

The existence of a solution of (C') is guaranteed by the results in [6]. Clearly, $u_2(x, t) > 0$ for $0 < x \leq 1$, $t > 0$ because $(u_2)_x(1, t) = 0$. Suppose that $u_{k+1} \geq u_k$ on D_T ; then set $\psi(x, t) = e^{\lambda x - 2\lambda^2 t}(u_{k+2} - u_{k+1})$ ($\lambda > 0$), on D_T ψ satisfies

$$\psi_t = \psi_{xx} + (-2\lambda + f'(u_{k+2}))\psi_x + [-\lambda^2 - \lambda f'(u_{k+2}) + f''(\theta_0)(u_{k+1})_x]\psi,$$

and at $x = 1$, $0 < t < T$,

$$\psi_x \geq (g_\xi(\theta_1, \overline{u_{k+1}}) + \lambda)(u_{k+1} - u_k) e^{\lambda - 2\lambda^2 t},$$

where θ_0 is between u_{k+2} and u_{k+1} and θ_1 is between u_{k+1} and u_k . We choose λ sufficiently large such that the coefficient of ψ is negative and $(g_\xi + \lambda)$ is positive. Then, since ψ cannot have a negative minimum inside and $\psi_x(1, t) \geq 0$, it follows that $\psi \geq 0$; i.e., $u_{k+2} \geq u_{k+1}$ on D_T . On the other hand, let $g_M = \sup_{|\xi| \geq M} |g(\xi, \eta)|$, since $u_1 = 0$, by induction and the preceding lemma, we can see that $u_k \leq M$, $k = 1, 2, \dots$, on D_{T_0} . Therefore, $u(x, t) = \lim_{k \rightarrow \infty} u_k(x, t)$ exists.

Using Theorem 3 in [12], we can show that u is a solution of (A) and possesses the necessary interior regularity. Uniqueness is guaranteed by the comparison principle.

Note that in constructing the positive solution, we did not use Theorem 3.1. For Burgers's equation, the two choices for g are therefore proper. The same argument allows us to extend the solution to $\bar{D}_{T+\delta}$ for some $\delta > 0$, if u remains bounded in $\bar{D}_T$. However, when the nonlinear terms and initial data satisfy certain conditions (as shown below), there are no global solutions for (A).

LEMMA 3.6. *Let $u(x, t)$ be a nonnegative solution of (A) with $u_x(x, 0) \geq 0$, $g(\xi, \eta) \geq 0$ for $\xi, \eta \geq 0$. Then $u_x(x, t) \geq 0$ in D_T . Additionally, if $f'(\xi) \geq 0$ for $\xi \geq 0$ and $g_\eta(\xi, \eta) \geq 0$ for $\xi \geq 0, \eta \geq 0$, then $u(x, t) \geq v(x, t)$, where v solves*

$$\begin{aligned}
 \text{(D)} \quad & v_t = v_{xx}, & 0 < x < 1, t > 0, \\
 & v(0, t) = 0, & t > 0, \\
 & v_x(1, t) = g(v, \bar{v}), & t > 0, \\
 & v(x, 0) \leq u(x, 0), & 0 \leq x \leq 1.
 \end{aligned}$$

Proof. Since u_x satisfies a linear equation with u_x nonnegative on the parabolic boundary, the first assertion follows. Then u is a supersolution of (D), and when the comparison theorem is applied, the second statement is valid.

THEOREM 3.7. Suppose that $f'(\xi) \geq 0$ for $\xi \geq 0$, and $g(\xi, \eta) \geq 0$, $g_\eta(\xi, \eta) \geq 0$ for $\xi, \eta \geq 0$, and

$$(p+1)G(\xi, \eta) \leq \xi g(\xi, \eta) \quad \text{for } \xi, \eta \geq 0, \quad (3.1)$$

where $G(\xi, \eta) = \int_0^\xi g(s, \eta) ds$, $p > 1$. Furthermore, let $v_0(x)$ satisfy

$$\begin{aligned}
 v_0(0) = 0, \quad v_0(x) \geq 0, \quad v_0''(x) + (f(v_0))' &\geq 0; \\
 v_0'(1) = g(v_0(1), \bar{v}_0), &
 \end{aligned} \quad (3.2)$$

$$\frac{1}{2} \int_0^1 (v_0'(x))^2 dx < G(v_0(1), \bar{v}_0); \quad (3.3)$$

then, if $u_0(x) \geq v_0(x)$ on $[0, 1]$, the solution of (A) blows up in finite time.

Proof. By the lemma, it is sufficient to show that $v(x, t)$ with $v(x, 0) = v_0(x)$ blows up. Note that in this case, we have $v_t \geq 0$ in D_T . Using a function similar to that in [11], we define

$$J(t) = \int_0^t \int_0^1 v^2(x, \eta) dx d\eta + (T^* - t) \int_0^1 v^2(x, 0) dx + \beta(t + \tau)^2,$$

where τ, β, T^* are positive constants with $T^* \geq T$:

$$\begin{aligned}
 J'(t) &= 2 \int_0^t \int_0^1 v v_\eta dx d\eta + 2\beta(t + \tau) \\
 &= -2 \int_0^t \int_0^1 v_x^2 dx d\eta + 2 \int_0^t v(1, \eta) g(v, \bar{v}) d\eta + 2\beta(t + \tau)
 \end{aligned}$$

and

$$J''(t) = -2 \int_0^1 v_x^2 dx + 2v(1, t) g(v, \bar{v}) + 2\beta.$$

Therefore,

$$\begin{aligned}
 J''(t) &= 4(\alpha + 1) \left[\int_0^t \int_0^1 v_\eta^2 dx d\eta + \beta \right] \\
 &\quad + 2 \left\{ \alpha \int_0^1 v_x^2 dx - (\alpha + 1) \int_0^1 v_0'^2 dx + v(1, t) g(v, \bar{v}) \right. \\
 &\quad \left. - 2(\alpha + 1) \int_0^t \frac{\partial}{\partial \eta} \left(\int_{v_0(1)}^{v(1, \eta)} g(s, \bar{v}) ds \right) d\eta \right. \\
 &\quad \left. + 2(\alpha + 1) \int_0^t \int_{v_0(1)}^{v(1, \eta)} g_v(s, \bar{v}) \bar{v}_\eta ds d\eta - (2\alpha + 1) \beta \right\} \\
 &\geq 4(\alpha + 1) \left[\int_0^t \int_0^1 v_\eta^2 dx d\eta + \beta \right] + 2 \left\{ [v(1, t) g(v, \bar{v}) - 2(\alpha + 1) G(v, \bar{v})] \right. \\
 &\quad \left. + 2(\alpha + 1) \left[G(v_0(1), \bar{v}_0) - \frac{1}{2} \int_0^1 v_0'^2 dx - \frac{(2\alpha + 1)}{2(\alpha + 1)} \beta \right] \right\}.
 \end{aligned}$$

Making use of (3.1) with $\alpha = (p - 1)/2$ and (3.3) with

$$\beta = \frac{2(\alpha + 1)}{(2\alpha + 1)} \left[G(v_0(1), \bar{v}_0) - \frac{1}{2} \int_0^1 v_0'^2 dx \right],$$

we obtain

$$J''(t) \geq 4(\alpha + 1) \left[\int_0^t \int_0^1 v_\eta^2 dx d\eta + \beta \right].$$

Hence,

$$\begin{aligned}
 JJ'' - (\alpha + 1)(J')^2 &\geq 4(\alpha + 1) \left\{ \left[\int_0^t \int_0^1 v^2 dx d\eta + \beta(t + \tau)^2 \right] \left[\int_0^t \int_0^1 v_\eta^2 dx d\eta + \beta \right] \right. \\
 &\quad \left. - \left[\int_0^t \int_0^1 vv_\eta dx d\eta + \beta(t + \tau) \right]^2 \right\} \geq 0.
 \end{aligned}$$

Consequently, $(J^{-\alpha})'' \leq 0$. Also, $(J^{-\alpha}(0))' < 0$. Therefore, $J^{-\alpha}(t)$ has a positive root and thus the conclusion follows. ■

For example, consider Burgers' equation with $g(\xi, \eta) = a\xi^p\eta^q$ ($p > 1$, $q > 0$) and $v_0(x) = Bx'$.

We discuss the case $1 < p < 3$ only. For $p \geq 3$, since $v_t \geq 0$, if $v_0(1) = B \geq 1$, the corresponding solution $v(x, t)$ is a supersolution of (A) with $1 < p < 3$, and hence by the comparison theorem, $v(x, t)$ dominates the one for $1 < p < 3$.

For (3.2), we must require

$$Br = \frac{aB^{p+q}}{(r+1)^q}, \quad B = \left[\frac{r(r+1)^q}{a} \right]^{1/(p+q-1)}. \quad (3.4)$$

From (3.3), $B^2 r^2 / (4r - 2) < aB^{p+q+1} / (p+1)(r+1)^q$, combining with (3.4), we have $(p+1)r^2 < r(4r-2)$; it follows that $r > 2/(3-p)$. Thus we choose $r > \max\{a, 2, 2/(3-p)\}$, which assures that $B \geq 1$.

For $g(\xi, \eta) = a\xi^p \eta^q - \frac{1}{2}\varepsilon \xi^2$ ($p \geq 2, q > 0$), we put u_0 so large that $a\bar{u}_0^q - \frac{1}{2}\varepsilon = b > 0$. Then u dominates the solution of (D) with $v_x(1, t) = bv^2(1, t)$. Following the similar procedure as above, we can show that v blows up and so does u .

Remark. 3.2. It is worth mentioning that all results in this section can be extended to a multidimensional analog of (A):

$$\begin{aligned} (P_n) \quad & u_t = \Delta u, & x \in \Omega, 0 < t < T, \\ & u(x, t) = 0, & x \in \sigma, 0 < t < T, \\ & (\partial u / \partial n)(x, t) = g(u(x, t), \bar{u}), & x \in \Sigma, 0 < t < T, \\ & u(x, 0) = u_0(x), & x \in \Omega. \end{aligned}$$

Here, g is as before, Ω is a domain in $\mathbb{R}^n$ ($n \geq 2$), $\text{cl}(\sigma \cup \Sigma) = \partial\Omega$, and $\sigma \cap \Sigma = \emptyset$.

For instance, the comparison theorem can be shown in the same manner as that of Theorem 3.1; and because the proof of local existence in Theorem 3.5 does not depend on the Green's function associated with the operator $L = \partial/\partial t - \partial^2/\partial x^2$, it can be used for (P_n) with one substitution $e^{\lambda\rho - \mu t} w$ for $e^{\lambda x - \mu t} w$, where ρ satisfies $\rho \geq 0$ in Ω , $\partial\rho/\partial n \geq 1$ on Σ . Such ρ was given in Lemma 6 of [10].

4. STABILITY AND INSTABILITY ANALYSIS

In this section, we shall study the long time behavior of global solutions. First, we state a lemma, whose proof is similar to that for Theorem 3.2A in [9] and, hence, is omitted.

LEMMA 4.1. *Let $u(x, t)$ be a bounded monotone solution of (A), then $u(x, t)$ exists for all time and tends to a stationary solution of (A).*

In order to obtain a complete result for stability and instability, we have to assume a few more conditions on the nonlinear terms f and g . This time, we replace f by εf ($\varepsilon \geq 0$); thus the stationary solution $w(x) = w(x, \varepsilon)$ can be treated as a function depending on the parameter ε .

THEOREM 4.2. *Suppose that f is twice continuously differentiable, $f'(\xi) > 0$ if $\xi > 0$, $g_\xi(\xi, \eta)$ is continuous for $\xi \geq 0$, $\eta \geq 0$, and $g_\eta(\xi, \eta) \geq 0$ for $\xi, \eta \geq 0$. Let $w(x, \varepsilon)$ be a continuously differentiable positive solution of (A₁) on some ε interval $[a, b]$ and let $w_1(\varepsilon) = w(1, \varepsilon)$. Then, if $w'_1(\varepsilon) \geq 0$ on $[a, b]$, the solutions are stable, whereas they are unstable if $w'_1(\varepsilon) < 0$.*

Proof. For the case $w'_1(\varepsilon) \geq 0$, set $v(x) = (\partial w / \partial \varepsilon)(x, \varepsilon)$, we claim that $v(x) > 0$ on $(0, 1)$ for $a \leq \varepsilon \leq b$. To see this, suppose that x_0 is a point in $(0, 1)$ such that $v(x_0) \leq 0$, $v'(x_0) = 0$. From the conservation law, we have

$$v' + \varepsilon f'(w)v = f(w(x_0)) - f(w(x)), \quad 0 < x < 1,$$

and

$$v(0) = 0, \quad v(1) \geq 0.$$

By using the formula for a linear equation, we find that

$$v(x_0) = e^{\int_{x_0}^1 \varepsilon f'(w(s)) ds} \left(v(1) + \int_{x_0}^1 (f(w(\tau)) - f(w(x_0))) e^{-\int_{x_0}^\tau \varepsilon f'(w(s)) ds} d\tau \right) > 0,$$

which is a contradiction.

From the above assertion, it follows that $w(x, \varepsilon_1) < w(x, \varepsilon_2)$ on $(0, 1)$ for $a \leq \varepsilon_1 < \varepsilon_2 \leq b$.

Let $u(x, t, \varepsilon_1)$ be a solution of (A) with $u_0(x, \varepsilon_1) = w(x, \varepsilon_2)$. Then, on $(0, 1)$, we have

$$\begin{aligned} u''_0 + \varepsilon_1 f'(u_0) u'_0 &= w_{xx}(x, \varepsilon_2) + \varepsilon_1 f'(w(x, \varepsilon_2)) w_x(x, \varepsilon_2) \\ &< w_{xx}(x, \varepsilon_2) + \varepsilon_2 f'(w(x, \varepsilon_2)) w_x(x, \varepsilon_2) \\ &= 0. \end{aligned}$$

Hence, recalling Corollary 3.3, we have $u_t \leq 0$ in D_T .

From the comparison theorem and monotonicity of u , we also have, on $(0, 1)$,

$$w(x, \varepsilon_1) \leq u(x, t, \varepsilon_1) \leq w(x, \varepsilon_2).$$

By Lemma 4.1, $\phi(x, \varepsilon_1) = \lim_{t \rightarrow \infty} u(x, t, \varepsilon_1)$ exists, and $w(x, \varepsilon_1) \leq \phi(x, \varepsilon_1) \leq w(x, \varepsilon_2)$. Letting $\varepsilon_2 \rightarrow \varepsilon_1^+$ yields $\phi(x, \varepsilon_1) \equiv w(x, \varepsilon_1)$, which shows that

$w(x, \varepsilon_1)$ is stable from above. With $\varepsilon_1 > \varepsilon_2$, in a similar manner, we can also prove that $w(x, \varepsilon_1)$ is stable from below.

If $w'_1(\varepsilon) < 0$, we know that $w(x, \varepsilon_2) < w(x, \varepsilon_1)$ in a subinterval $[x_1, 1]$ for $a \leq \varepsilon_1 < \varepsilon_2 \leq b$. Now let $u(x, t, \varepsilon_2)$ be a solution of (A) with $u_0(x, \varepsilon_2) = w(x, \varepsilon_1)$. Then, on $(0, 1)$, we find that

$$\begin{aligned} u''_0 + \varepsilon_2 f'(u_0) u'_0 &= w_{xx}(x, \varepsilon_1) + \varepsilon_2 f'(w(x, \varepsilon_1)) w_x(x, \varepsilon_1) \\ &> w_{xx}(x, \varepsilon_1) + \varepsilon_1 f'(w(x, \varepsilon_1)) w_x(x, \varepsilon_1) \\ &= 0. \end{aligned}$$

Therefore, $u_t \geq 0$ in D_T . Consequently, $u(x, t, \varepsilon_2)$ is increasing in t , which indicates that $w(x, \varepsilon_2)$ is unstable from above. Similarly, with $\varepsilon_1 > \varepsilon_2$, we can show that $w(x, \varepsilon_2)$ is unstable from below.

In particular, with the help of the above theorem, we are able to establish crucial criteria for stability and to give the bifurcation diagrams for Burgers' equation with $f(\xi) = \frac{1}{2}\varepsilon\xi^2$ and $g(\xi, \eta) = a\xi^p\eta^q$ or $g(\xi, \eta) = a\xi^p\eta^q - \frac{1}{2}\varepsilon\xi^2$.

For problem (A₁) with $g(\xi, \eta) = a\xi^p\eta^q$, combining (2.3) and (2.4) yields

$$\varepsilon = w_1^{-1}(\varepsilon) \gamma^{-1} \ln \left(\frac{\gamma+1}{\gamma-1} \right). \quad (4.1)$$

By replacing ε in (2.3) with (4.1), we obtain

$$\begin{aligned} [w_1(\varepsilon)]^{(p+q-1)/(1+q)} \\ = \left(\frac{1}{2a} \right)^{1/(1+q)} (\gamma^2 - 1)^{1/(1+q)} \gamma^{-1} \ln \left(\frac{\gamma+1}{\gamma-1} \right) \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^{-q/(1+q)} \end{aligned} \quad (4.2)$$

First, we discuss the case $p+q \neq 1$; (4.2) can be written in the form of

$$[w_1(\varepsilon)]^{(p+q-1)/(1+q)} = \left(\frac{2^q}{a} \right)^{1/(1+q)} \Phi(\gamma),$$

where $\Phi(\gamma)$ is the same as in (2.8) with $\lambda = 1/(1+q)$, $\mu = q/(1+q)$. Hence,

$$\left(\frac{p+q-1}{1+q} \right) [w_1(\varepsilon)]^{(p-2)/(1+q)} w'_1(\varepsilon) = \left(\frac{2^q}{a} \right)^{1/(1+q)} \Phi'(\gamma) \gamma'(\varepsilon). \quad (4.3)$$

Since $\Phi'(\gamma) > 0$ on $(1, \infty)$, if $p+q > 1$, $w'_1(\varepsilon)$ has the same sign as $\gamma'(\varepsilon)$, whereas it has the opposite sign if $p+q < 1$.

If $p = 2$, from (2.3), we have

$$\varepsilon^{1+q} = \varphi(\gamma) = 2a(\gamma^2 - 1)^{-1} \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^q.$$

It is clear that $\gamma'(\varepsilon) < 0$ because $\varphi'(\gamma) < 0$. Hence, when $p = 2$, $w(x, \varepsilon)$ is an unstable branch. For $p \neq 2$, recalling the equation $\Phi(\gamma) = \delta$ in (2.7), we find that

$$\Phi'(\gamma) \gamma'(\varepsilon) = \left(\frac{p+q-1}{p-2} \right) 2^{(1-p)/(p-2)} a^{1/(2-p)} \varepsilon^{(1+q)/(p-2)}.$$

If $p > 2$, we have that $\gamma'(\varepsilon) < 0$ because $\Phi'(\gamma) < 0$. The same holds for $1 \leq p < 2$ and $0 < p < 1$ ($p+q > 1$) because $\Phi'(\gamma) > 0$ and $(p+q-1)/(p-2) < 0$.

Next, consider that $p+q < 1$. We observe that $\Phi'(\gamma)$ is first positive and then negative; it follows that there are two branches, one satisfying $\gamma'(\varepsilon) > 0$ and another satisfying $\gamma'(\varepsilon) < 0$.

Finally, for $p+q = 1$, because $\Phi'(\gamma) > 0$ on $(1, \infty)$, it is clear from (4.3) that $\gamma'(\varepsilon) = 0$. Using (4.1), we can see that $w_1'(\varepsilon) < 0$.

Therefore, if $p \geq 1$ or if $0 < p < 1$ but $p+q \geq 1$, $w(x, \varepsilon)$ is unstable, whereas there are two branches of solutions for $p+q < 1$ —one unstable, the other stable.

Now, we briefly discuss $w_1(\varepsilon)$ graphically and draw the corresponding diagrams. We first attempt to study the relation between ε and γ based on (2.7), and then describe the behavior of $w_1(\varepsilon)$.

If $1 \leq p \neq 2$, or $0 < p < 1$ but $p+q > 1$, $\varepsilon \rightarrow 0$ as $\gamma \rightarrow \infty$ and $\varepsilon \rightarrow +\infty$ as $\gamma \rightarrow 1^+$; thus, $w_1(0) = (2^q/a)^{1/(p+q-1)}$ and $w_1(+\infty) = +\infty$. The same is true for $p = 2$ because $\varepsilon^{1+q} = \varphi(\gamma)$.

For $p+q = 1$, there is a unique solution $w(x, \varepsilon)$ if $a < 2^{1-p}$. Since ε is independent of γ , by (4.1), we have that $w_1(0) \rightarrow \infty$ and $w_1(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow +\infty$.

For $p+q < 1$, ε ranges over a finite interval $[0, \varepsilon]$ because $0 \leq \Phi(\gamma) \leq M$. Then the stable branch of $w(x, \varepsilon)$ increases from $(2^q/a)^{1/(p+q-1)}$ to $w_1(\varepsilon_1)$ and the unstable one grows from $w_1(\varepsilon_1)$ to infinity as $\varepsilon \rightarrow 0^+$. The graphs are shown in Figs. 1.1 and 1.2.

Using a similar argument, we can show the corresponding diagrams for problem (A₁) with $g(\xi, \eta) = a\xi^p\eta^q - \frac{1}{2}\varepsilon\xi^2$. Note that the equation for ε is the same as (4.1). By replacing ε in (2.9) with (4.1), we obtain

$$[w_1(\varepsilon)]^{(p+q-1)/(1+q)} = \left(\frac{1}{2a} \right)^{1/(1+q)} \gamma^{(1+q)/(1+q)} \ln \left(\frac{\gamma+1}{\gamma-1} \right) \left[\ln \left(\frac{\gamma^2}{\gamma^2-1} \right) \right]^{-q/(1+q)}. \quad (4.4)$$

First, we discuss the case $p + q \neq 1$; (4.4) can be written in the form of

$$[w_1(\varepsilon)]^{(p+q-1)/(1+q)} = \left(\frac{2^q}{a}\right)^{1/(1+q)} \Phi(\gamma),$$

where $\Phi(\gamma)$ is the same as in (2.11) with $\lambda = 1/(1+q)$, $\mu = q/(1+q)$. Hence,

$$\left(\frac{p+q-1}{1+q}\right) [w_1(\varepsilon)]^{(p-2)/(1+q)} w_1'(\varepsilon) = \left(\frac{2^q}{a}\right)^{1/(1+q)} \Phi'(\gamma) \gamma'(\varepsilon). \quad (4.5)$$

Since $\Phi'(\gamma) < 0$ on $(1, \infty)$, if $p + q < 1$, $w_1'(\varepsilon)$ has the same sign as $\gamma'(\varepsilon)$, whereas it has the opposite sign if $p + q > 1$.

If $p = 2$, from (2.9), we have

$$\varepsilon^{1+q} = \varphi(\gamma) = 2a\gamma^{-2} \left[\ln \left(\frac{\gamma^2}{\gamma^2 - 1} \right) \right]^q.$$

It is clear that $\gamma'(\varepsilon) < 0$ because $\varphi'(\gamma) < 0$.

For $p \neq 2$, recalling the equation $\Phi(\gamma) = \delta$ in (2.10), we find that

$$\Phi'(\gamma) \gamma'(\varepsilon) = \left(\frac{p+q-1}{p-2} \right) 2^{(1-p)/(p-2)} a^{1/(2-p)} \varepsilon^{(1+q)/(p-2)}.$$

If $p > 2$, we have that $\gamma'(\varepsilon) < 0$ because $\Phi'(\gamma) < 0$. The same holds for $p < 2$ ($p + q \geq 2$) because $\Phi'(\gamma) > 0$ and $(p+q-1)/(p-2) < 0$, and for $p + q < 1$ since $\Phi'(\gamma) < 0$.

Next, consider that $1 < p + q < 2$. We observe that $\Phi'(\gamma)$ is first negative and then positive; it follows that there are two branches, one satisfying $\gamma'(\varepsilon) > 0$ and another satisfying $\gamma'(\varepsilon) < 0$.

Finally, for $p + q = 1$, because $\Phi'(\gamma) < 0$ on $(1, \infty)$, it is clear from (4.5) that $\gamma'(\varepsilon) = 0$. Using (4.1), we can see that $w_1'(\varepsilon) < 0$. Therefore, if $p + q \geq 2$, $w(x, \varepsilon)$ is stable; if $p + q \leq 1$, $w(x, \varepsilon)$ is unstable, whereas there are two branches of solutions for $1 < p + q < 2$ —one unstable, the other stable. The graphs are shown in Figs. 2.1 and 2.2.

At the end of this section, we investigate the stability of stationary solutions of Burgers' equation with $g(\xi, \eta) = a\xi^p\eta^q$ or $g(\xi, \eta) = a\xi^p\eta^q - \frac{1}{2}\varepsilon\xi^2$ ($a < 0$). The positive solutions of (A) can be obtained by a contraction method, as in [9].

First, consider problem (A) with $g(\xi, \eta) = \xi^2$. For convenience, we denote its solution and its stationary solution by $u_A(x, t)$ and $w_A(x)$, respectively. By the comparison principle, it follows that the solution $u(x, t)$ of (A) with $g(\xi, \eta) = a\xi^p\eta^q$ or $g(\xi, \eta) = a\xi^p\eta^q - \frac{1}{2}\varepsilon u^2$ satisfies: if $0 \leq u(x, 0) \leq u_A(x, 0)$, then $0 \leq u(x, t) \leq u_A(x, t)$. Choosing $u_A(x, 0) =$

$(1 - \delta)w_A(x)$ and following a process similar to that in proving Theorem 3.2A of [9], we can see that $u_A(x, t) \rightarrow 0$ as $t \rightarrow \infty$, which implies that the null solution of (A_1) is stable from above.

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